

习题课第二章、

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1. 解: $X \in \{0, 1, 2\}$.

$$P(X=0) = \frac{C_8^2}{C_{10}^2} = \frac{28}{45}$$

$$P(X=1) = \frac{C_2^1 \cdot C_8^1}{C_{10}^2} = \frac{16}{45}$$

$$P(X=2) = 1 - P(X=0) - P(X=1) = \frac{1}{45}.$$

4. 解: $\sum_{k=1}^{\infty} f(k) = 1$ 且 $f(k) > 0$.

$$\sum_{k=1}^{\infty} f(k) = C \cdot \sum_{k=1}^{\infty} \frac{\lambda^k}{k!}$$

$$\left(e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} \right)$$

$$= C \cdot \left(\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} - 1 \right)$$

$$= C \cdot (e^{\lambda} - 1) = 1$$

⇓

$$C = \frac{1}{e^{\lambda} - 1}$$

5. 解: 1° $x < 1$ 时, $F(x) = 0$

2° $x \geq 3$ 时, $F(x) = 1$.

3° $1 \leq x < 2$ 时, $F(x) = 0.2$

4° $2 \leq x < 3$ 时, $F(x) = 0.2 + 0.3 = 0.5$

综上所述:

$$F(x) = \begin{cases} 0, & x < 1 \\ 0.2, & 1 \leq x < 2 \\ 0.5, & 2 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$

11. 解: (1). $F(x)$ 在 $x=1$ 处右连续.

$$\lim_{x \rightarrow 1^+} F(x) = F(1) = k \cdot 1^2 = k.$$

$$\lim_{x \rightarrow 1^+} F(x) = \lim_{x \rightarrow 1^+} 1 = 1$$

$$\therefore k = 1.$$

$$\begin{aligned} (2) \quad P(0.25 < X < 0.75) &= F(0.75) - F(0.25) \\ &= 0.5 \end{aligned}$$

$$(3) \quad f(x) = \begin{cases} 0, & x < 0 \\ 2x, & 0 \leq x \leq 1 \\ 0, & x > 1 \end{cases}$$

$\Downarrow$

$$f(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{其他} \end{cases}$$

$$14) \text{ 解: } p = C_4^3 \times 0.5^3 \times 0.5^1 = 0.25$$

$$17. \text{ 解: (1) 由 } \int_{-\infty}^{+\infty} f(x) dx = 1 \text{ 知.}$$

$$\int_{-1}^1 \frac{k}{\sqrt{1-x^2}} dx = 1$$

$$\therefore k \cdot \arcsin x \Big|_{-1}^1 = 1 \Rightarrow k = \pi^{-1}$$

$$(2) P(|X| < \frac{1}{2})$$

$$= \int_{-\frac{1}{2}}^{\frac{1}{2}} f(x) dx = \frac{1}{\pi} \int_{-1/2}^{1/2} \frac{1}{\sqrt{1-x^2}} dx$$

$$= 1/3.$$

13) 解: 1° $x \leq -1$ 时, $F(x) = 0$

2° $x \geq 1$ 时, $F(x) = 1$

3° $-1 < x < 1$ 时,

$$\underline{F(x)} = \int_{-\infty}^x f(t) dt$$

$$= \int_{-1}^x \frac{1}{\pi} \cdot \frac{1}{\sqrt{1-t^2}} dt$$

$$= \frac{1}{2} + \frac{\arcsin x}{\pi}$$

综上所述:

$$F(x) = \begin{cases} 0, & x \leq -1 \\ \frac{1}{2} + \frac{1}{\pi} \arcsin x, & -1 < x < 1 \\ 1, & x \geq 1 \end{cases}$$

21. 解: $X \sim N(0, 1)$, $Y = |X|$

$\therefore Y \geq 0$

1° $y \leq 0$ 时, $F_Y(y) = 0$.

$$\begin{aligned}
 2^\circ \quad y > 0 \text{ 时} \quad F_Y(y) &= P(Y \leq y) \\
 &= P(|X| \leq y) \\
 &= P(-y \leq X \leq y) \\
 &= \Phi(y) - \Phi(-y) \\
 &= 2\Phi(y) - 1
 \end{aligned}$$

(注: $\Phi(x) + \Phi(-x) = 1$)

$\therefore y > 0$ 时, $f_Y(y) = F_Y'(y) = 2\phi(y)$.

综上所述:

$$f_Y(y) = \begin{cases} \sqrt{\frac{2}{\pi}} \cdot e^{-\frac{y^2}{2}}, & y > 0 \\ 0, & y \leq 0. \end{cases}$$

22. 解: $g(x) = \frac{1}{x} \quad (y = \frac{1}{x} \Rightarrow x = \frac{1}{y})$.
 $h(y) = \frac{1}{y}$

$$Y = \frac{1}{X} \in (1, +\infty)$$

当 $y > 1$ 时, $f_Y(y) = f_X(h(y)) \cdot |h'(y)|$
 $= 3 \cdot (\frac{1}{y})^2 \cdot |-\frac{1}{y^2}|$

$$= 3 \cdot y^{-4}$$

$$\therefore f_Y(y) = \begin{cases} 3 \cdot y^{-4}, & y > 1 \\ 0, & y \leq 1. \end{cases}$$

25. 解: $X \sim \text{Exp}(\frac{1}{2})$.

$$\therefore f_X(x) = \begin{cases} 2 \cdot e^{-2x}, & x > 0 \\ 0, & x \leq 0. \end{cases}$$

$$Y = 1 - e^{-2X} \in (0, 1)$$

$y \in (0, 1)$ 时, $f_Y(y) = 0$.

$$y \in (0, 1) \text{ 时, } f_Y(y) = \underbrace{f_X(h(y))} \cdot \underbrace{|h'(y)|}$$

$$Y = 1 - e^{-2X}$$

$$e^{-2X} = 1 - Y$$

$$-2X = \log(1 - Y)$$

$$X = -\frac{1}{2} \log(1 - Y)$$

$$\therefore \underbrace{h(y)} = -\frac{1}{2} \log(1 - y)$$

30. 解: (1)

$$F(x, y) = \int_{-\infty}^y \int_{-\infty}^x \underline{f(u, v)} du dv$$

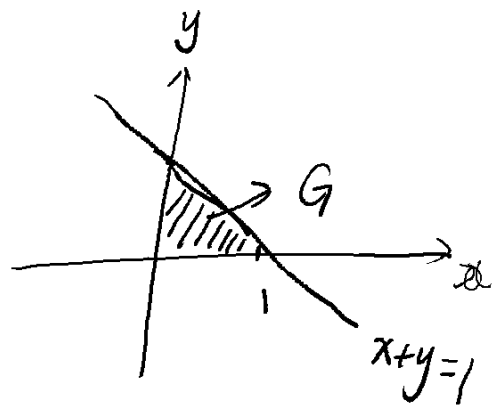
$$= \begin{cases} \int_0^y \int_0^x e^{-(u+v)} du dv, & x > 0, y > 0 \\ 0, & \text{其他} \end{cases}$$

$$\therefore F(x, y) = \begin{cases} (1 - e^{-x})(1 - e^{-y}), & x > 0, y > 0 \\ 0, & \text{其他} \end{cases}$$

(2). $P((X, Y) \in G)$

$$= \iint_G f(x, y) dx dy$$

$$= \int_0^1 \int_0^{1-x} e^{-(x+y)} dy dx$$



$$= 1 - 2e^{-1}$$

31. 解: (1) 由 $\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy = 1$ 知.

$$\int_0^{+\infty} \int_0^{+\infty} A \cdot e^{-(3x+4y)} dx dy = 1$$

$\Downarrow$

$$A = 12.$$

(2) 解: $f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy$

$$= \begin{cases} \int_0^{+\infty} 12 e^{-(3x+4y)} dy, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

$$= \begin{cases} 3 \cdot e^{-3x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

$$\begin{aligned}
 \text{同理 } f_Y(y) &= \int_{-\infty}^{+\infty} \underline{f(x, y)} dx \\
 &= \begin{cases} 4 \cdot e^{-4y}, & y > 0 \\ 0, & y \leq 0. \end{cases}
 \end{aligned}$$

$$(3) \text{ 解: } P(0 \leq X \leq 1, 0 \leq Y \leq 2)$$

$$= \int_0^2 \int_0^1 f(x, y) dx dy$$

$$= \int_0^2 \int_0^1 12 \cdot e^{-(3x+4y)} dx dy$$

$$= (1 - e^{-3})(1 - e^{-8}).$$

34. 解:

$$(1) f_X(x) = \begin{cases} 5, & 0 \leq x \leq 0.2 \\ 0, & \text{其他} \end{cases}$$

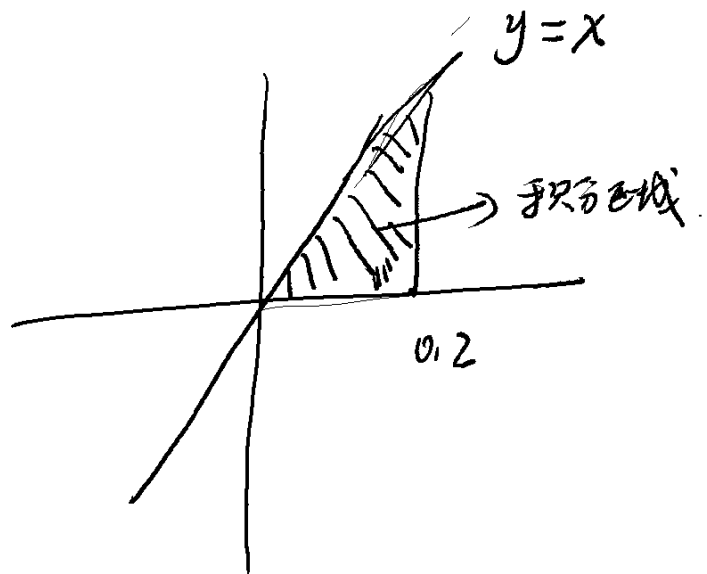
$$f_Y(y) = \begin{cases} 5 \cdot e^{-5y}, & y > 0 \\ 0, & y \leq 0 \end{cases}$$

$$f(x, y) = f_X(x) \cdot f_Y(y).$$

$$\therefore f(x, y) = \begin{cases} \frac{25 \cdot e^{-5y}}{1}, & \underline{0 \leq x \leq 0.2}, \underline{y > 0} \\ 0, & \text{其他} \end{cases}$$

$$(2) \text{ 解: } P(Y \leq X)$$

$$= \iint_{y \leq x} f(x, y) dx dy$$



$$= \int_0^{0.2} \int_0^x 25 \cdot e^{-5y} dy dx$$

$$= e^{-1}$$

35. 解: $P(X, Y \text{ 至少有一个小于 } 1/2)$

$$= 1 - P(X > \frac{1}{2}, Y > \frac{1}{2})$$

$$= 1 - \int_{\frac{1}{2}}^{+\infty} \int_{\frac{1}{2}}^{+\infty} f(x, y) dx dy$$

$$= 1 - \int_{\frac{1}{2}}^2 \int_{\frac{1}{2}}^1 \frac{1}{2} dx dy$$

$$= 1 - 3/8 = 5/8.$$

38. 解:

$$F_X(x) = F(x, +\infty)$$

$$= \lim_{y \rightarrow +\infty} F(x, y)$$

$$= \begin{cases} 0 & , x < 0, \\ \frac{x^2}{1+x^2} & , x > 0 \end{cases}$$

$$\underline{F_Y(y)} = F(+\infty, y)$$

$$= \lim_{x \rightarrow +\infty} F(x, y) = \begin{cases} 1, & y > 1 \\ y^3, & 0 < y \leq 1 \\ 0, & y \leq 0 \end{cases}$$

$$F(x, y) = F_X(x) \cdot F_Y(y)$$

$\therefore X, Y$ 相互独立.

40. 解:

$$f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy$$

$$= \begin{cases} \int_0^1 (x+y) dy, & 0 \leq x \leq 1 \\ 0 & , \text{其他.} \end{cases}$$

$$= \begin{cases} \frac{1}{2} + x, & 0 \leq x \leq 1 \\ 0 & , \text{其他.} \end{cases}$$

同理得

$$f_Y(y) = \begin{cases} \frac{1}{2} + y, & 0 \leq y \leq 1 \\ 0, & \text{其他} \end{cases}$$

$x=0.4, y=0.4$ 时, $f(x, y) \neq f_X(x) \cdot f_Y(y)$

$\therefore X, Y$ 不独立.

42. 解: $f_z(z) = \underline{f_x * f_y(z)}$

$$= \underline{\int_{-\infty}^{+\infty} f_x(z-y) \cdot \underline{f_y(y)} dy.}$$

$$\underline{f_y(y)} = \begin{cases} \frac{1}{2b}, & -b \leq y \leq b \\ 0, & \text{其他.} \end{cases}$$

$$\underline{f_x(x)} = \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{(x-a)^2}{2\sigma^2}}.$$

$$\therefore f_z(z)$$

$$= \int_{-b}^b \frac{1}{2b} \cdot \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{(z-y-a)^2}{2\sigma^2}} dy$$

$$= \frac{1}{2b} \cdot \int_{-b}^b \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{(y+a-z)^2}{2\sigma^2}} dy$$

$$= \frac{1}{2b} \int_{-b}^b \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{(y-(z-a))^2}{2\sigma^2}} dy$$

($N(z-a, \sigma^2)$ 落在 $(-b, b)$ 内概率)

($X \sim N(\mu, \sigma^2)$ 落在 (x_1, x_2) 内的概率为:

$$P(x_1 < X \leq x_2)$$

$$= \Phi\left(\frac{x_2 - \mu}{\sigma}\right) - \Phi\left(\frac{x_1 - \mu}{\sigma}\right)$$

$$= \frac{1}{2b} \left\{ \Phi\left(\frac{b - (z-a)}{\sigma}\right) \right.$$

$$\left. - \Phi\left(\frac{-b - (z-a)}{\sigma}\right) \right\}$$